



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

EDMS 17-300-1

TECHNICAL SPECIFICATION

FOR

LOW VOLTAGE CURRENT TRANSFORMERS

Issue: MAR-2021 / Rev- 1

FOR: network needs / low voltage panels and meter panels*

يتم العمل بهذا المواصفة اعتباراً من تاريخ اعتمادها باستثناء اختبار تيار
القصر يتم أخذه في الاعتبار ابتداءً من تاريخ ٢٠٢١-٩-١

❖ توجد بنود اختيارية (* Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

CONTENTS

<u>Clause</u>	<u>Description</u>	<u>Page</u>
1	SCOPE	3
2	APPLICABLE CODES AND STANDARDS	3
3	ENVIRONMENTAL CONDITIONS	3
4	DESIGN AND CONSTRUCTION	5
5	MARKING	5
6	SEALING	6
7	TESTS	6
8	SAMPLE	6
9	GUARANTEE	6
10	GUARANTEE SCHEDULE	8
11	DELIVERABLES	8

LIST OF TABLES

<u>Clause</u>	<u>Description</u>	<u>Page</u>
-	TABLE NO 1	3
-	TABLE NO 2	4
-	TABLE NO 3	5
-	TABLE NO 4	5
-	TABLE NO 5	5
-	TABLE OF GUARANTEE	9



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

1. SCOPE:

This specification covers the minimum technical requirements for design, engineering, manufacturing, testing, inspection and performance of low voltage current transformers used in the LV distribution system which is Solid earthed neutral with rated voltage 400/230V & rated frequency 50 HZ.

2. Applicable standards:

Unless otherwise specified in this specification, the low voltage CTs should comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance with the applicable IEC standards as following table:

Table (1)

S. No	Standard No.	Description
1	IEC 61869-1	Instrument transformers general requirements
2	IEC 61869-2	Additional requirements for current transformers.

3. ENVIRONMENTAL CONDITIONS:

The performance of CTs should be guaranteed for the following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer:



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

Table (2)

Minimum ambient temperature	-5°c
Maximum ambient temperature	45°c (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m

4. DESIGN AND CONSTRUCTION:

- 4.1. Rated Primary and Secondary Current of LV CTs
 - a) The rated secondary current is 5A.
 - b) The rated primary currents as given at table (3).
- 4.2. The secondary wires should be high-temperature insulated copper and not less than 2.5 mm²
- 4.3. The connection of the secondary wire and the winding should be rigidly and permanently fixed, and the opening to where the secondary wire passes through should be provided with wire grommets or better to provide mechanical protection on the wire.
- 4.4. The LV CTs should be of ring - type /window-type without hinges and suitable for horizontal and vertical installation.
- 4.5. The LV CTs should be provided with a suitable tin or silver plated electrolytic copper terminal block with purity not less than 99.99%.
- 4.6. It should be suitable for use in narrow switching cubicles and suitable for threading over cables /bus bar size according to the following tables:



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

Table (3) for network needs

CT Ratio	100/ 5	150/ 5	200/5	300/ 5	400/ 5	500/ 5	600/ 5	800/5	1000/ 5	1500/ 5	1600/ 5	2000/5	2500/5
Bus bar Size mm*mm Tolerance +30 %	10*40	10*40	10*40	10*40	10*60	10*80	10*80	10*80 2*10*40	10*100 2*10*50	10*120 2*10*60	10*120 2*10*60	10*120 2*10*60	2*10*100
Cable Diameter mm Tolerance +30 %	26	26	26	26	44	55	62	70	70	-	-	-	-
MIN Burden (VA)	2.5	2.5	10	10	10	10	10	10	10	10	10	10	10
MIN Short circuit (KA)	6	9	12	18	24	30	36	48	50	50	50	50	50

Note: The max outer dimension is related to the bus bar clearance 2 cm for CT rates up to 600/5 and 4 cm for CT rates greater than 600/5.

Table (4)* optional asEDC requirements for new (low voltage panels and meter panels)

CT Ratio	100 /5	150 /5	200/ 5	300 /5	400 /5	500 /5	600 /5	800/ 5	1000 /5	1500 /5	1600 /5	2000/ 5	2500/ 5	3000 /5	4000 /5	5000/ 5
Bus bar Size mm*mm	10*30	10*30	5*40 10*20	10*30	10*40	10*40	10*50	10*80 2*10*40	10*100 2*10*50	10*120 2*10*60	10*120 2*10*60	10*120 2*10*60	2*10*100	2*10*100	2*10*120 3*10*80	3*10*100
MIN Burden (VA)	2.5	2.5	10	10	10	10	10	10	10	10	10	10	10	10	10	10
MIN Short circuit (KA)	6	9	12	18	24	30	36	48	50	50	50	50	50	50	50	50

Note: The max outer dimension is related to the bus bar clearance 2 cm for CT rates up to 600/5 and 4 cm for CT rates greater than 600/5.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

Table (5)* optional asEDC requirements for both new and network needs

CT Ratio	1000/5	2000/5	3000/5	4000/5
LV bushing C.S.A mm ²	63*12	100*15	120*15	120*20
MIN Short circuit (KA)	50	50	50	50

4.1. The C.T. should be single phase, for metering purpose and should have the following technical data:

- It should be insulated by cast resin or suitable insulation.
- Maximum service voltage 600V.
- C.T. Accuracy class 0.5S (class 0.2 **Optional as per ... EDC Requirements**).
- The rated short time thermal current (r.m.s) should be according to table (3, 4, and 5) with a minimum duration of 1 second.
- Rated power-frequency withstand voltage (r.m.s) for 1 min should be 3 KV.
- Safety Factor for Measuring: FS5 (FS10 **Optional as per ... EDC Requirements**)
- Rated continuous current: 1.2 I rated primary current.
- Insulation class $\geq$ E (temperature rise limit "K" 70 °C @ 45°C ambient) and according to IEC 61869-2.
- Rated dynamic current I_p (peak): 2.5x I thermal.

5. MARKING:

The following data should be marked on CT itself or laser printed on the CT, and located in a place such that they are visible and legible when CT is installed:

- The manufacturer's name/ model
- The serial number and Year of Manufacturing
- Class of insulation



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

- Symbol ofEDC.
- The rated primary and secondary current
- The polarization on both sides.
- The rated frequency
- The rated output, Burden and corresponding accuracy class

6. SEALING:

CT should be provided with suitable tool for sealing the secondary terminals of the current transformer.

7. TESTS:

a. Type tests:

- Temperature-rise test
- Tests for accuracy
- Short-Time current tests (short circuit test for the highest rate for each models (series)).

Type tests should be in accordance with IEC 61869-1, 2.

Type test certificates should be submitted with the offer.

b. Routine/Acceptance Tests:

- Power-frequency voltage withstand tests
- Tests for accuracy
- Verification of markings
- Turns ratio error.

Acceptance tests should be performed on random samples and should be selected and witnessed by Distribution Company representative.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

Routine tests should be performed in compliance with IEC61869-1,2

8. Sample:

The tenderer should provide a sample C.T. with his offer. The sample should correspond in all respects with the material they intend to supply.

9. GUARANTEE:

The tenderer should guarantee the CT against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, the good quality and manufacturing of the materials should be guaranteed.

10. GUARANTEE TABLES:

Tenders not accompanied with attached table of guarantee fulfilled and signed by tenderers should not be considered.

11. SUBMITTALS:

- a) The tenderer should submit a complete set of dimensional drawings, technical catalogues and user manual showing the CT type offered and its performance in full details.
- b) A copy of the latest applicable High Voltage Research Center Test Certificate should be attached with the tender, and the rest of the type tests must be completed from an accredited and neutral laboratory in accordance with IEC standards.
- c) A valid approved letter must be submitted from the Egyptian Electricity Holding Company attached to the submitted offer.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-300-1

Date: 02-03-2021

TABLE OF GUARANTEE

for CT/5A

. Standards to which C.Ts are manufactured and tested		
. Name of Manufacturer		
. Place of manufacturing		
. Year of manufacturing		
. Product type		
. C.T ratio		
. Maximum service voltage	V	
. Accuracy class		
. Rated burden	VA	
. Temperature rise limit of winding	°C	
. Rated continuous current	x In	
. Safety Factor for Measuring		
. Thermal class of insulation		
. Rated short time thermal current (r.m.s)	KA	
. Dimensions(H x W x D)		
. Weight	Kg	
. Max. dimensions - that C.T can be threaded over:-		
. For busbars	mm*mm	
. For cable	mm*diam.	
. For bushing	mm*mm	

We guarantee the above data for our offered C.T. toEDC.

Signature

Date: / /